<u>So far</u>: Convergence tests

⓪ · $a_n \not\to 0$ as $n \to \infty \Rightarrow \sum a_n$ diverges

· telescoping $\Rightarrow$ compute s_n, find $\lim s_n$

① Comparison test

② Dyadic criterion

③ Root test

④ Ratio test

⑤ Dirichlet's criterion

] Today

<u>Thm</u> (Ratio test) Let $\sum_{n=1}^{\infty} a_n$ be a series of nonzero numbers.

① If $\limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$, then $\sum_{n=1}^{\infty} a_n$ converges absolutely.

② If $\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| > 1$, then $\sum_{n=1}^{\infty} a_n$ diverges.

<u>Rmk</u> If $\liminf_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| \le 1 \le \limsup_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$, then the ratio test is inconclusive. E.g. both $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$ satisfy $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = 1$.

<u>Pf</u>: By Cesàro–Stolz,

$$\liminf \left|\frac{a_{n+1}}{a_n}\right| \le \liminf |a_n|^{\frac{1}{n}} \le \limsup |a_n|^{\frac{1}{n}} \le \limsup \left|\frac{a_{n+1}}{a_n}\right|.$$

① $\limsup \left|\frac{a_{n+1}}{a_n}\right| < 1 \Rightarrow \limsup |a_n|^{\frac{1}{n}} < 1$

$\Rightarrow \sum a_n$ converges absolutely by the root test.

② $\liminf \left|\frac{a_{n+1}}{a_n}\right| > 1 \Rightarrow \liminf |a_n|^{\frac{1}{n}} > 1$

$\Rightarrow \sum a_n$ diverges by the root test. $\square$

<u>Ex</u> $\sum_{n=1}^{\infty} \frac{n}{3^n}$

$$a_n = \frac{n}{3^n} \Rightarrow \left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)\,3^n}{n\,3^{n+1}} = \frac{n+1}{3n} \xrightarrow{n\to\infty} \frac{1}{3}$$

We have $\limsup \left|\frac{a_{n+1}}{a_n}\right| = \lim \left|\frac{a_{n+1}}{a_n}\right| = \frac{1}{3} < 1$, and so the series converges by the ratio test.

<u>Ex</u> $\frac{1}{2} + \frac{1}{3} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^3} + \dots$

The sequence $\left|\frac{a_{n+1}}{a_n}\right|$ is:

$$\frac{1/3}{1/2} = \frac{2}{3}, \quad \frac{1/2^2}{1/3} = \frac{1}{2}\cdot\frac{3}{2}, \quad \frac{1/3^2}{1/2^2} = \left(\frac{2}{3}\right)^2, \quad \frac{1/2^3}{1/3^2} = \frac{1}{2}\left(\frac{3}{2}\right)^2, \dots$$

Note that

$$\left(\frac{2}{3}\right)^k \to 0 \quad \text{as } k\to\infty$$

$$\frac{1}{2}\left(\frac{3}{2}\right)^k \to +\infty \quad \text{as } k\to\infty$$

Thus $\liminf \left|\frac{a_{n+1}}{a_n}\right| = 0$, $\limsup \left|\frac{a_{n+1}}{a_n}\right| = +\infty$, and so the ratio test is inconclusive.

The sequence $|a_n|^{\frac{1}{n}}$ is:

$$\frac{1}{2}, \quad \frac{1}{3^{1/2}}, \quad \frac{1}{2^{2/3}}, \quad \frac{1}{3^{2/4}} = \frac{1}{\sqrt{3}}, \quad \frac{1}{2^{3/5}}, \quad \frac{1}{3^{3/6}} = \frac{1}{\sqrt{3}}, \dots$$

Note that

$$\frac{1}{\sqrt{3}} \longrightarrow \frac{1}{\sqrt{3}}$$

$$\frac{1}{2^{\frac{k}{2k-1}}} = \frac{1}{2^{\frac{1}{2}}\cdot 2^{\frac{1}{4k-2}}} \longrightarrow \frac{1}{\sqrt{2}} \quad \text{as } k\to\infty$$

Thus $\limsup_{n\to\infty} |a_n|^{\frac{1}{n}} = \frac{1}{\sqrt{2}} < 1$, and so the series

converges by the root test.

<u>Thm</u> (Dirichlet's criterion) Let $\{a_n\}_{n\geq 1}$, $\{b_n\}_{n\geq 1}$ be sequences. If $\{a_n\}_{n\geq 1}$ is decreasing, $\lim_{n\to\infty} a_n = 0$, and the sequence $\{\sum_{k=1}^{n} b_k\}_{n\geq 1}$ is bounded, then $\sum_{n=1}^{\infty} a_n b_n$ converges.

<u>Pf</u>: For $n\geq 1$, set $t_n = \sum_{k=1}^{n} b_k$. As $\{t_n\}_{n\geq 1}$ is bounded, $\exists M>0$ s.t.

$$|t_n| \leq M \quad \forall n\geq 1.$$

We will prove $\sum a_n b_n$ converges using the Cauchy criterion. Fix $\varepsilon>0$. As $\lim_{n\to\infty} a_n = 0$, $\exists N$ s.t.

$$|a_n| < \frac{\varepsilon}{2M} \quad \forall n\geq N.$$

Then, for $n\geq m\geq N$ we have

$$\begin{aligned}
\sum_{k=m}^{n} a_k b_k &= \sum_{k=m}^{n} a_k (t_k - t_{k-1}) \\
&= \sum_{k=m}^{n} a_k t_k - \sum_{k=m}^{n} a_k t_{k-1} \qquad \text{Set } \ell = k-1 \\
&= \sum_{\ell=m}^{n} a_\ell t_\ell - \sum_{\ell=m-1}^{n-1} a_{\ell+1} t_\ell \qquad \ell = k \\
&= -\sum_{\ell=m}^{n} (a_{\ell+1} - a_\ell) t_\ell + a_{n+1} t_n - a_m t_{m-1}
\end{aligned} \quad \star$$

$$\Rightarrow \left|\sum_{k=m}^{n} a_k b_k\right| \leq \sum_{\ell=m}^{n} \underbrace{|a_{\ell+1} - a_\ell|}\, |t_\ell| + |a_{n+1}||t_n| + |a_m||t_{m-1}|$$

$= a_\ell - a_{\ell+1} \geq 0$, since $\{a_n\}$ decreasing

$$\leq \sum_{\ell=m}^{n} (a_\ell - a_{\ell+1}) \cdot M + a_{n+1} \cdot M + a_m \cdot M$$

$$= (a_m - \cancel{a_{n+1}}) M + \cancel{a_{n+1} \cdot M} + a_m \cdot M$$

$$= 2Ma_m < \varepsilon$$

So $\sum a_n b_n$ converges by the Cauchy criterion. $\square$

<u>Rmk</u> The identity ⊛ is sometimes called "summation by parts".

<u>Cor</u> (Alternating series test) If $\{a_n\}_{n\geq 1}$ is a decreasing sequence and $\lim_{n\to\infty} a_n = 0$, then $\sum_{n=1}^{\infty} (-1)^n a_n$ converges.

<u>Pf</u>: Apply Dirichlet's criterion to $b_n = (-1)^n$. Note that the sequence

$$b_1 + b_2 + \ldots + b_n = \begin{cases} -1 & n \text{ odd} \\ 0 & n \text{ even} \end{cases}$$

is bounded. $\square$

<u>Ex</u> $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$

- Dyadic criterion does not apply: $a_n \not\geq 0 \ \forall n$
- Comparison test does not work: $\sum \frac{1}{\sqrt{n}}$ diverges
- Ratio and root tests are inconclusive

This series converges by the alternating series test, since $\left\{\frac{1}{\sqrt{n}}\right\}_{n\geq 1}$ is decreasing and $\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$.